

Multi-Band Patch Antenna Array for Out-of-Band Aided Millimeter Wave Communication

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Abstract—Future wireless communication systems will integrate both sub-6 GHz and millimeter wave (mmWave) frequency bands within multi-antenna architectures to meet the increasing demand for high data rates. In such multi-band systems, reliable information obtained from the sub-6 GHz band can be exploited to support communication at mmWave frequencies. To ensure that both systems experience similar multi-path propagation effects, the sub-6 GHz and mmWave antenna arrays have to be co-located and precisely aligned. However, such a configuration may adversely alter the radiation characteristics of the arrays, potentially degrading their performance. In this paper, we investigate the impact of positioning a mmWave antenna structure in front of a sub-6 GHz antenna structure. Through both simulations and measurements, we evaluate how the presence of the mmWave structure affects the radiation pattern of the sub-6 GHz one. The results demonstrate that the influence of the mmWave structure on the sub-6 GHz performance is minor, indicating that co-located configurations are feasible with negligible degradation.

Index Terms—antenna array, multi-band, mmWave, sub-6 GHz, out-of-band information.

I. INTRODUCTION

Due to the high utilization of conventional sub-6 GHz frequency bands, the available bandwidth is limited, which constrains data transmission rates. In contrast, millimeter wave (mmWave) frequency bands (24 GHz – 300 GHz) [1] offer significantly wider bandwidths, making them well-suited to support the growing demand for high data rates [2]. Furthermore, mmWave systems are increasingly being deployed in conjunction with sub-6 GHz systems to enable multi-band communication and improve reliability [3], [4].

Compared to mmWave bands, which suffer from high propagation losses, sub-6 GHz bands exhibit more favorable propagation characteristics, resulting in more robust link reliability. This makes them a valuable source of out-of-band information that can be leveraged to support communication at mmWave frequencies. So far, numerous methods that exploit sub-6 GHz out-of-band information to aid mmWave communication have been proposed [5]–[8]. These methods employ co-located and aligned sub-6 GHz and mmWave antenna arrays (sharing the same central point) to ensure that both systems experience similar multi-path propagation effects.

However, the design of such co-located and aligned multi-band antenna arrays poses significant practical challenges. A common implementation involves placing the mmWave array in front of the sub-6 GHz array. While this configuration allows

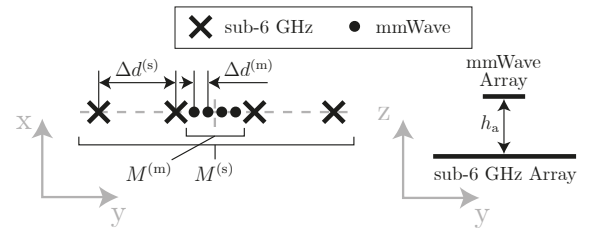


Fig. 1. The multi-band uniform linear array (ULA) consists of sub-6 GHz and mmWave antenna arrays that are co-located and precisely aligned (sharing the same geometric center) and separated by an inter-array spacing of h_a .

for spatial alignment, it can adversely affect the radiation pattern of the sub-6 GHz array, potentially degrading its performance. Therefore, it is crucial to evaluate the performance of such multi-band antenna structures under realistic hardware constraints.

Contribution: In this paper, we investigate the impact of positioning a mmWave antenna structure in front of a sub-6 GHz structure. We first assess, through both simulations and measurements, how the presence of the mmWave array influences the radiation pattern of a single sub-6 GHz element. The analysis is then extended to full sub-6 GHz and mmWave arrays using simulations.

Organization: Section II describes the multi-band antenna array design used in this paper. Simulation and measurement results are presented in Section III. Finally, Section IV concludes the paper.

Notation: The superscript $(\cdot)^{(b)}$ stands for frequency-band dependent values, where $b \in \{s, m\}$. In this context, s stands for the sub-6 GHz frequency band and m stands for the mmWave frequency band.

II. MULTI-BAND ANTENNA ARRAY DESIGN

We consider a multi-band antenna array composed of co-located sub-6 GHz and mmWave arrays, operating simultaneously in the radiative far-field regime. The sub-6 GHz array comprises $M^{(s)}$ elements, while the mmWave array consists of $M^{(m)}$ elements (see Fig. 1). The arrays are precisely aligned, sharing the same geometric center, with an inter-array spacing of h_a , as illustrated in Fig. 1. The sub-6 GHz and mmWave arrays are implemented as uniform linear arrays (ULAs) of patch antenna elements. In both arrays, adjacent antenna elements are mutually separated by $\Delta d^{(b)} = 0.5 \lambda^{(b)}$, where

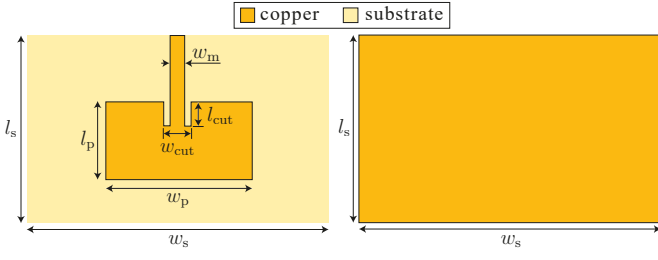


Fig. 2. On the top layer (left), each patch element, with dimensions $l_p^{(b)} \times w_p^{(b)}$, is fed by a microstrip line of width $w_m^{(b)}$ and features an inset feed of dimensions $l_{cut}^{(b)} \times w_{cut}^{(b)}$. The bottom layer (right) consists of a solid copper ground plane.

TABLE I
SIMULATION PARAMETERS

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
Substrate Type	RT-Duroid 5880	RO4350B
Relative Dielectric Constant ϵ_r	2.2	3.66
Substrate Thickness h_s	1.5 mm	0.254 mm
Copper Thickness h_c	70 μ m	70 μ m
Dissipation Factor $\tan \delta$	0.0004	0.0037
Microstrip Line Width w_m	4.53307 mm	0.504685 mm
Patch Element Width w_p	46.47 mm	3.851 mm
Patch Element Length l_p	38.89 mm	2.985 mm
Inset Feed Cut Width w_{cut}	5.7899 mm	0.5276 mm
Inset Feed Cut Length l_{cut}	10 mm	1.005 mm

$\lambda^{(b)}$ denotes the wavelength corresponding to the respective frequency band. Each patch antenna element has dimensions $l_p^{(b)} \times w_p^{(b)}$ and is fed by a microstrip line of width $w_m^{(b)}$ with inset feed dimensions $l_{cut}^{(b)}$ and $w_{cut}^{(b)}$. Fig. 2 shows top and bottom layers of a single patch antenna. The patch elements are designed to resonate at the carrier frequency $f_c^{(b)}$ of their respective systems. The patch elements are etched on a substrate of thickness $h_s^{(b)}$ with copper thickness $h_c^{(b)}$, relative dielectric constant $\epsilon_r^{(b)}$ and dissipation factor $\tan \delta^{(b)}$. The total array dimensions are given by $l_s^{(b)} \times w_s^{(b)}$, where $l_s^{(b)} = \Delta d^{(b)} + \lambda^{(b)}/2$ and $w_s^{(b)} = M^{(b)}\Delta d^{(b)} + \lambda^{(b)}/2$. The specific parameter values for each frequency band are summarized in Tab. I.

Moreover, we validate the accuracy of our antenna design in terms of return loss through simulations in ANSYS HFSS [9]. For the sub-6 GHz array, the antenna resonates at 2.55 GHz with a return loss of 18.47 dB, whereas the mmWave array resonates at 25.5 GHz with a return loss of 13.16 dB. These results demonstrate satisfactory impedance matching at the target frequencies in both frequency bands. Furthermore, we complement the simulations results with experimental measurements. A single sub-6 GHz patch element is manufactured in-house according to the design parameters listed in Tab. I.

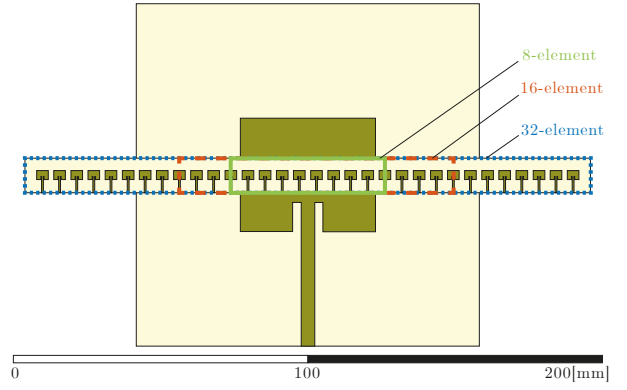


Fig. 3. A mmWave ULA with variable size (8-element, 16-element or 32-element) is positioned in front of the single sub-6 GHz patch antenna.

We measure the return loss of the fabricated patch antenna using a vector network analyzer (VNA) (Rohde&Schwarz ZVA8) under controlled laboratory conditions. The measured results show that the antenna resonates at 2.5625 GHz with a return loss of 18.22 dB, corresponding to only a 0.5% deviation from the target frequency of 2.55 GHz. This small shift can be attributed to fabrication tolerances and substrate parameter variations, which are common in practical antenna prototyping. At the exact target frequency of 2.55 GHz, the measured return loss is 12.96 dB, which still indicates satisfying impedance matching and acceptable performance for practical operation.

III. SIMULATION AND MEASUREMENT RESULTS

In this section, we investigate the effect of placing a mmWave antenna structure in front of a sub-6 GHz antenna structure. Section III-A focuses on a single sub-6 GHz antenna element, where the influence of the mmWave array is assessed through both simulations and measurements. Section III-B extends the analysis to full antenna arrays, considering ULAs using simulations. Consistent with common practice, the mmWave arrays are modeled with a larger number of elements compared to their sub-6 GHz counterparts [10].

A. Single Antenna Element Analysis

We first analyze the impact of placing a mmWave ULA with different dimensions (i.e., number of antenna elements) in front of a single sub-6 GHz patch antenna element. Specifically, for the mmWave array, we consider $M^{(m)} > 1$, while for the sub-6 GHz antenna we have $M^{(s)} = 1$. The analysis is carried out using both full-wave simulations and antenna measurements.

The simulations are performed at 2.55 GHz using the commercial software ANSYS HFSS [9]. The simulated structure with an inter-array spacing of $h_a = 10$ mm is shown in Fig. 3. We show in Fig. 4 the realized gain patterns of the sub-6 GHz element for different mmWave array configurations in the E-plane and H-plane. The results for the realized gain in the main-lobe direction and the half-power beamwidth (HPBW) are summarized in Tab. II. As a baseline reference, we use the realized gain of the standalone sub-6 GHz patch

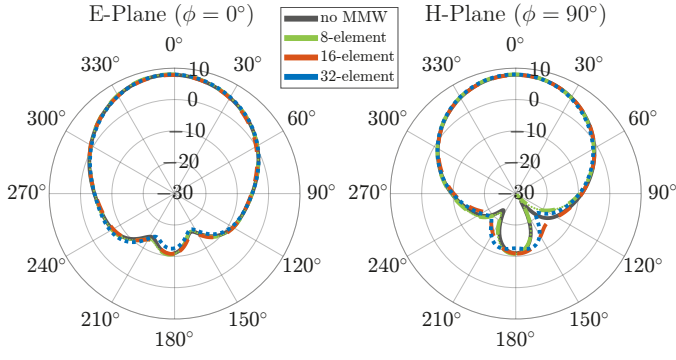


Fig. 4. The realized gain of the sub-6 GHz patch element in the main lobe direction is approximately 7.96 dBi, with a negligible degradation of around 0.12 dB when the mmWave array is present.

TABLE II
RESULTS

		mmWave Array Configuration	Realized Gain	HPBW	
				E-plane	H-plane
Single Antenna	Simulated	no mmWave	7.96 dBi	68°	64°
		8-element	7.86 dBi	68°	62°
		16-element	7.84 dBi	68°	64°
		32-element	7.86 dBi	68°	62°
	Measured	no mmWave	7.20 dBi	75°	74°
		8-element	6.95 dBi	71°	74°
		16-element	7.10 dBi	75°	74°
Array	Simulated	no mmWave	15.33 dBi	68°	12°
		8-element	15.23 dBi	68°	12°
		16-element	15.20 dBi	66°	12°
		32-element	15.18 dBi	68°	12°

element without any mmWave structure present, which is approximately 7.96 dBi.

The results demonstrate that the presence of the mmWave array has only a negligible effect on the radiation performance of the sub-6 GHz patch. In particular, the maximum observed degradation in realized gain is around 0.12 dB. The HPBW in the E-plane remains unchanged at 68°, while in the H-plane it decreases slightly from 64° to 62° (a reduction of 3%). The results indicate that the electromagnetic interaction between the two structures is minimal at sub-6 GHz frequencies.

Furthermore, the simulation results are complemented by field-pattern measurements conducted in the anechoic chamber at TU Wien. The measurement setup, shown in Fig. 5, employs a horn antenna as the probe antenna and the in-house manufactured sub-6 GHz patch antenna as the antenna under test. To emulate the presence of a mmWave array in front of the patch, unetched printed circuit boards with copper cladding on both sides are used. These boards, fabricated on the mmWave substrate with the thickness specified in Tab. I, have dimensions corresponding to 8-element and 16-element mmWave arrays. The mmWave printed circuit boards are placed with an inter-array spacing of $h_a = 10$ mm from the sub-6 GHz patch, with a ROHACELL layer in between. The

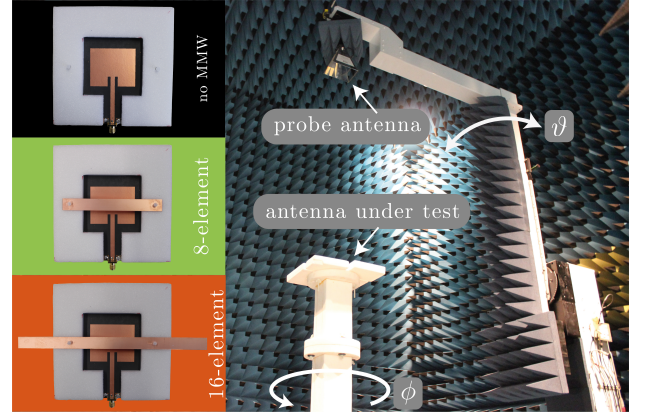


Fig. 5. Measurement setup for field-pattern measurements conducted in the anechoic chamber at TU Wien.

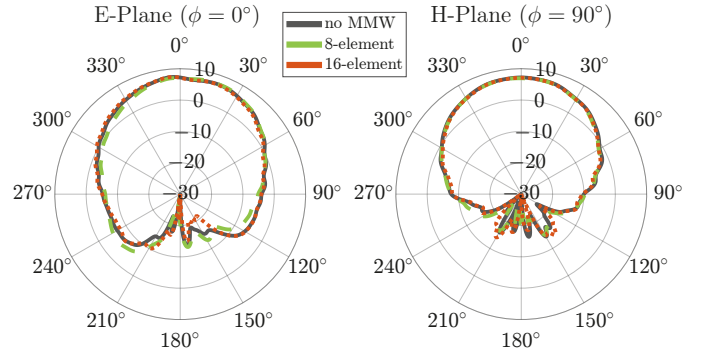


Fig. 6. The measured realized gain of the sub-6 GHz patch element in the main lobe direction is approximately 7.2 dBi, with a degradation of around 0.25 dB when the mmWave array is present.

realized gain of the single sub-6 GHz patch element is then measured at 2.55 GHz.

We show the measured gain patterns of the sub-6 GHz element for different mmWave array configurations in the E-plane and H-plane in Fig. 6. As a baseline, we measure the realized gain of the standalone sub-6 GHz patch element without mmWave printed circuit boards present. The realized gain of the baseline configuration is approximately 7.2 dBi in the main-lobe direction. The corresponding measured HPBW is 75° in the E-plane and 74° in the H-plane. Consistent with the simulation analysis, the measurements confirm that the presence of the mmWave array has only a negligible effect on the radiation performance of the sub-6 GHz patch. As shown in Tab. II, the maximum observed degradation in realized gain is about 0.25 dB in the main-lobe direction. The HPBW in the E-plane decreases slightly from 75° to 71° (a 5% reduction), while in the H-plane it remains unchanged at 74°.

B. Antenna Array Analysis

In this analysis, we investigate the impact of placing mmWave arrays of varying size in front of sub-6 GHz arrays through simulations. We evaluate the radiation performance of the sub-6 GHz arrays at 2.55 GHz using the commercial software ANSYS HFSS [9]. The simulated structure, with

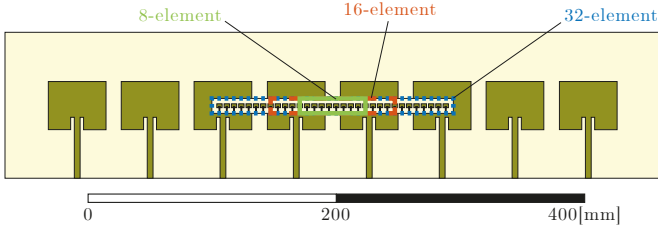


Fig. 7. A mmWave ULA with variable size (8-element, 16-element or 32-element) is positioned in front of an 8-element sub-6 GHz ULA.

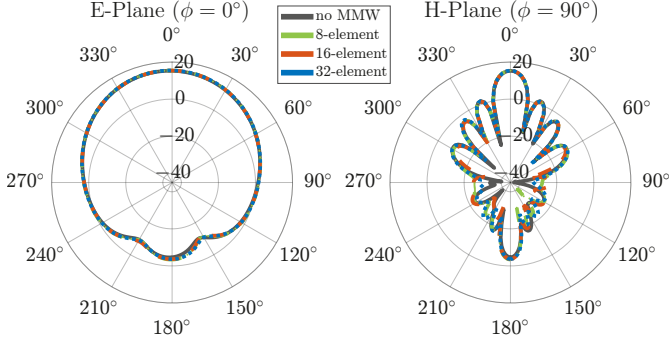


Fig. 8. The realized gain of the 8-element sub-6 GHz ULA is approximately 15.33 dBi in the main lobe direction, with only a negligible reduction of about 0.15 dB due to the presence of the mmWave array.

an inter-array spacing of $h_a = 10$ mm, is shown in Fig. 7. In Fig. 8, we show the realized gain patterns of the 8-element sub-6 GHz array in the E-plane and H-plane.

As a baseline, we employ the realized gain of the 8-element sub-6 GHz ULA without the mmWave structure, which is approximately 15.33 dBi in the main-lobe direction. As shown in Tab. II, the presence of the mmWave array introduces only a negligible effect: the maximum reduction in realized gain is about 0.15 dB in the main-lobe direction, the E-plane HPBW decreases slightly from 68° to 66° (3%) and the H-plane HPBW remains unchanged at 12° .

Finally, we analyze the reverse scenario, i.e., the effect of a sub-6 GHz array on the radiation performance of a mmWave array. For this case, as shown in Fig. 9, we simulate the realized gain of an 8-element mmWave ULA at 25.5 GHz using the HFSS. The realized gain is only marginally reduced, from 15.16 dBi in the baseline configuration without the sub-6 GHz array to 15.08 dBi when the sub-6 GHz array is present. The HPBW remains unchanged at 64° in the E-plane and 12° in the H-plane. Furthermore, the results indicate that the sub-6 GHz ULA effectively acts as an additional ground plane for the mmWave array, causing negligible impact on the main-lobe gain while suppressing the back lobe. Specifically, the back-lobe level decreases from -3.4 dBi to -7.5 dBi, thereby improving the radiation performance of the mmWave array.

IV. CONCLUSION

Both simulation and measurement results confirm that the presence of the mmWave array has only a minor impact on the radiation characteristics of both a single sub-6 GHz patch

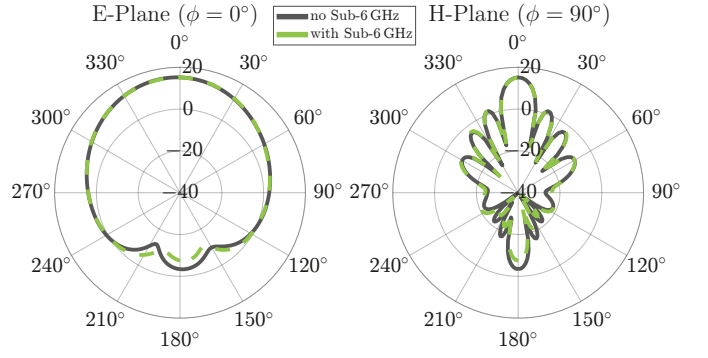


Fig. 9. The sub-6 GHz ULA acts as an additional ground plane for the mmWave ULA, resulting in negligible impact on the main-lobe gain while effectively suppressing the back lobe.

element and a sub-6 GHz ULA. Similarly, the sub-6 GHz array has a negligible influence on the main-lobe performance of the mmWave array, while beneficially reducing back-lobe radiation. These findings demonstrate that co-located sub-6 GHz and mmWave ULA configurations are feasible and introduce only negligible performance degradation. Nevertheless, future work should extend this analysis to uniform planar array configurations to assess radiation effects in two-dimensional antenna structures.

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